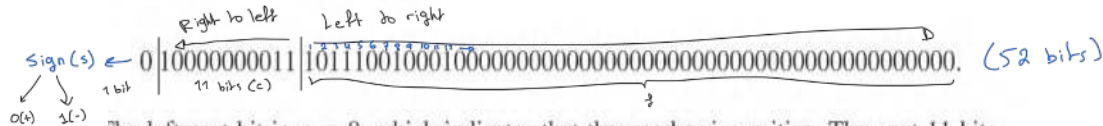


1.2

Convert to Floating-number

$$\begin{aligned} (-1)^s &= 1 & (-1)^c &= -1 \\ (s) \text{ pos} & & (c) \text{ neg} & \end{aligned}$$

$$(-1)^s 2^{c-1023} (1+f)$$



The leftmost bit is $s = 0$, which indicates that the number is positive. The next 11 bits, 0000000011, give the characteristic and are equivalent to the decimal number

$$c = 1 \cdot 2^{10} + 0 \cdot 2^9 + \dots + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 1024 + 2 + 1 = 1027.$$

$$f = 1 \cdot \left(\frac{1}{2}\right)^1 + 1 \cdot \left(\frac{1}{2}\right)^3 + 1 \cdot \left(\frac{1}{2}\right)^4 + 1 \cdot \left(\frac{1}{2}\right)^5 + 1 \cdot \left(\frac{1}{2}\right)^8 + 1 \cdot \left(\frac{1}{2}\right)^{12}.$$

As a consequence, this machine number precisely represents the decimal number

$$\begin{aligned} (-1)^s 2^{c-1023} (1+f) &= (-1)^0 \cdot 2^{1027-1023} \left(1 + \left(\frac{1}{2} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{256} + \frac{1}{4096}\right)\right) \\ &= 27.56640625. \end{aligned}$$

To calculate c , take the base as 2

$$10000000011_2 = 1 \cdot 2^{10} + 1 \cdot 2^1 + 1 \cdot 2^0 = 2^{10} + 2^1 + 2^0 = 1027$$

To calculate f , take the base as $\frac{1}{2}$

$$10111001000100000000000000000000000000000000_2 = \left(\frac{1}{2}\right)^1 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^5 + \left(\frac{1}{2}\right)^8 + \left(\frac{1}{2}\right)^{12} = \frac{2961}{4096}$$

$$2^{c-1023} (1+f) = 2^{1027-1023} \left(1 + \frac{2961}{4096}\right) = 27.56640625$$

Actual value = p Approximate value = p^* $p - p^* = \text{actual error (A.E)}$
Absolute error = $|p - p^*|$ Relative error = $\frac{\text{Absolute error}}{|p|}$

Suppose that p^* is an approximation to p . The **actual error** is $p - p^*$, the **absolute error** is $|p - p^*|$, and the **relative error** is $\frac{|p - p^*|}{|p|}$, provided that $p \neq 0$.

at most $\leq$
at least $\geq$
more than $>$
less than $<$

EXERCISE SET 1.2

$$\textcircled{1} \textcircled{4} \text{ A.E.} = |p - p^*| = \pi - \frac{22}{7} = 0.0012644$$

$$\text{R.E.} = \frac{|p - p^*|}{|p|} = 0.0004$$

$$\textcircled{3} \textcircled{4} \text{ R.E.} \leq 10^{-3}$$

$$\frac{|p - p^*|}{|p|} \leq 10^{-3}$$

$$\frac{|150 - p^*|}{150} \leq 10^{-3}$$

$$\frac{|150 - p^*|}{150} \leq 10^{-3}$$

$$|150 - p^*| \leq 10^{-3} \times 150$$

$$-10^{-3} \times 150 \leq 150 - p^* \leq 10^{-3} \times 150$$

$$-150 \quad -150 \quad -150$$

$$-150 - 150 \times 10^{-3} \leq p^* \leq -150 + 150 \times 10^{-3}$$

$$-150 - 150 \times 10^{-3} \leq p^* \leq -150 + 150 \times 10^{-3}$$

$$-150.15 \leq p^* \leq -149.85$$

$$149.85 \leq p^* \leq 150.15 \quad \text{belongs to } p^* \in [149.85, 150.15]$$

1. Compute the absolute error and relative error in approximations of
- p
- by
- p^*
- .

a. $p = \pi, p^* = 22/7$

b. $p = \pi, p^* = 3.1416$

c. $p = e, p^* = 2.718$

d. $p = \sqrt{2}, p^* = 1.414$

2. Compute the absolute error and relative error in approximations of
- p
- by
- p^*
- .

a. $p = e^{10}, p^* = 22000$

b. $p = 10^\pi, p^* = 1400$

c. $p = 8!, p^* = 39900$

d. $p = 9!, p^* = \sqrt{18\pi}(9/e)^9$

3. Suppose
- p^*
- must approximate
- p
- with relative error at most
- 10^{-3}
- . Find the largest interval in which
- p^*
- must lie for each value of
- p
- .
-
- Less than, greater than*

a. $150 = p$

b. $900 = p$

c. $1500 = p$

d. $90 = p$

4. Find the largest interval in which
- p^*
- must lie to approximate
- p
- with relative error at most
- 10^{-4}
- for each value of
- p
- .

a. π

b. e

c. $\sqrt{2}$

d. $\sqrt[3]{7}$

5. Perform the following computations (i) exactly, (ii) using three-digit chopping arithmetic, and (iii) using three-digit rounding arithmetic. (iv) Compute the relative errors in parts (ii) and (iii).

a. $\frac{4}{5} + \frac{1}{3}$

b. $\frac{4}{5} \cdot \frac{1}{3}$

c. $\left(\frac{1}{3} - \frac{3}{11}\right) + \frac{3}{20}$

d. $\left(\frac{1}{3} + \frac{3}{11}\right) - \frac{3}{20}$

$$\textcircled{4} \textcircled{b} \text{ R.E.} \leq 10^{-4}$$

$$\frac{|p - p^*|}{|p|} \leq 10^{-4}$$

$$\frac{|e - p^*|}{e} \leq 10^{-4}$$

$$\frac{|e - p^*|}{e} \leq 10^{-4}$$

$$|e - p^*| \leq 10^{-4} \cdot e$$

$$-e \cdot 10^{-4} \leq e - p^* \leq e \cdot 10^{-4}$$

$$-e \quad -e \quad -e$$

$$-e \cdot 10^{-4} \leq p^* \leq -e \cdot 10^{-4}$$

$$-2.71855 \leq p^* \leq -2.71801$$

$$2.71801 \leq p^* \leq 2.71855$$

$$p \in [2.71801, 2.71855]$$

$$\textcircled{5} \text{ a) i) } \frac{4}{5} + \frac{1}{3} = \frac{17}{15}$$

$$\text{ii) } \frac{4}{5} = 0.800$$

Under a non zero, just add 2 zeros

$$\frac{1}{3} = 0.333$$

$$\frac{4}{5} + \frac{1}{3} = 0.800 + 0.333 = 1.13 \text{ (3 digits)}$$

If the decimal place is 0.08

then the answer will be 0.0800 (for 3 digits)

leading zero isn't counted

$$\text{iii) } \frac{4}{5} = 0.800$$

$$\frac{1}{3} = 0.333$$

$$\frac{4}{5} + \frac{1}{3} = 0.800 + 0.333 = 1.13 \text{ (3 digits)}$$

1.3

★ You have to apply Taylor's theory

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

6. Find the rates of convergence of the following sequences as $n \rightarrow \infty$.

a. $\lim_{n \rightarrow \infty} \sin \frac{1}{n} = 0$

b. $\lim_{n \rightarrow \infty} \sin \frac{1}{n^2} = 0$

c. $\lim_{n \rightarrow \infty} \left(\sin \frac{1}{n} \right)^2 = 0$

d. $\lim_{n \rightarrow \infty} [\ln(n+1) - \ln(n)] = 0$

7. Find the rates of convergence of the following functions as $h \rightarrow 0$.

a. $\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1$

b. $\lim_{h \rightarrow 0} \frac{1 - \cos h}{h} = 0$

c. $\lim_{h \rightarrow 0} \frac{\sin h - h \cos h}{h} = 0$

d. $\lim_{h \rightarrow 0} \frac{1 - e^h}{h} = -1$

① $\lim_{h \rightarrow 0} \frac{1 - e^h}{h} = \frac{1 - e^0}{0} = \frac{0}{0}$

$$= \lim_{h \rightarrow 0} \frac{1 - (1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + \dots)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-(h + \frac{h^2}{2!} + \frac{h^3}{3!} + \dots)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-1 - \frac{h}{2!} - \frac{h^2}{3!} - \dots}{1} = -1 \quad O(h)$$

⑦ ① $\frac{\sin 0}{0} = \frac{0}{0}$ (indeterminate form)

$$\lim_{h \rightarrow 0} \frac{(h - \frac{h^3}{3!} + \frac{h^5}{5!} - \dots)}{h} = \lim_{h \rightarrow 0} \frac{h(1 - \frac{h^2}{3!} + \frac{h^4}{5!} - \dots)}{h} = \lim_{h \rightarrow 0} (1 - \frac{h^2}{3!} + \frac{h^4}{5!} - \dots) = 1$$

$O(h^2)$

order of convergence

② $\lim_{h \rightarrow 0} \frac{1 - \cos h}{h} = \frac{1 - \cos 0}{0} = \frac{0}{0}$

$$\lim_{h \rightarrow 0} \frac{1 - (1 - \frac{h^2}{2!} + \frac{h^4}{4!} - \dots)}{h} = \lim_{h \rightarrow 0} \frac{(\frac{h^2}{2!} - \frac{h^4}{4!} + \dots)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(\frac{h}{2!} - \frac{h^3}{4!} + \dots)}{h} = \lim_{h \rightarrow 0} (\frac{h}{2!} - \frac{h^3}{4!} + \dots) = 0$$

$O(h)$

order of convergence

6. Find the rates of convergence of the following sequences as $n \rightarrow \infty$. $\rightarrow$ the limit goes to infinity, the limit does not exist

a. $\lim_{n \rightarrow \infty} \sin \frac{1}{n} = 0$ $\frac{1}{n} = 0$

b. $\lim_{n \rightarrow \infty} \sin \frac{1}{n^2} = 0$

c. $\lim_{n \rightarrow \infty} \left(\sin \frac{1}{n} \right)^2 = 0$

d. $\lim_{n \rightarrow \infty} [\ln(n+1) - \ln(n)] = 0$

~~A~~ $\sin n \leq n$

$-\frac{1}{n} \leq \sin \frac{1}{n} \leq \frac{1}{n}$

~~B~~ $\sin \frac{1}{n} \leq \frac{1}{n}$

(a) $\lim_{n \rightarrow \infty} \left(\sin \frac{1}{n} \right) = 0$

$\sin \frac{1}{n} \leq \frac{1}{n}$

$\lim_{n \rightarrow \infty} \sin \frac{1}{n} \leq \lim_{n \rightarrow \infty} \frac{1}{n} = 0$

$\Rightarrow \lim_{n \rightarrow \infty} \left(\sin \frac{1}{n} \right) = 0$

$O\left(\frac{1}{n}\right)$

(c) $\lim_{n \rightarrow \infty} \left(\sin \frac{1}{n} \right)^2 = 0$

$\sin \frac{1}{n} \leq \frac{1}{n}$

$\left(\sin \frac{1}{n} \right)^2 \leq \left(\frac{1}{n} \right)^2$

$\left(\sin \frac{1}{n} \right)^2 \leq \frac{1}{n^2}$

$\lim_{n \rightarrow \infty} \left(\sin \frac{1}{n} \right)^2 \leq \lim_{n \rightarrow \infty} \frac{1}{n^2} = 0$

$\Rightarrow \lim_{n \rightarrow \infty} \left(\sin \frac{1}{n} \right)^2 = 0$

$O\left(\frac{1}{n^2}\right)$

(b) $\lim_{n \rightarrow \infty} \sin \frac{1}{n^2} = 0$